What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Primary Myelofibrosis

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Adapted from: Primary Myelofibrosis

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Learning Objectives (4) · Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of primary myelofibrosis.
2. Explain the pathophysiology of primary myelofibrosis.
3. Interpret clinical findings to properly diagnose primary myelofibrosis.
4. Outline the [illegible] osis, including the rationale [illegible]

CASE CONNECTION

VB is a 68-year-old male who comes to your clinic with left-sided abdominal pain and fatigue. "I've been putting off coming in, but my wife insisted that I see you. The pain has been going on for several months, and it is getting worse. Actually, it's not a pain as much as a sort of a heaviness. It's wearing me out." VB's vital signs are normal, but on physical exam, you appreciate a massively enlarged spleen, with the tip at the pelvic brim. On his complete blood count (CBC), his WBC count is elevated. The report on the peripheral smear notes teardrop-shaped cells (also called dacryocytes).

What will be the next step in confirming this diagnosis? What will you tell VB about the condition? Consider your answers as you read, and we'll revisit VB at the end of the brick.

GO TO CONCLUSION ↓

What Is Primary Myelofibrosis?

Primary myelofibrosis (PMF) is a hematopoietic neoplasm in which the bone marrow gradually becomes filled with fibrosis. It is one among a group of disorders called **myeloproliferative neoplasms (MPNs)** and is the one with the worst prognosis.

As with all the myeloproliferative neoplasms, PMF is a **slow, chronic, progressive** disease, [illegible] rogression seen in acute hematopoietic malignancies such as acute myeloid leukemia.

Differentiation From Other Myeloproliferative Neoplasms

Primary myelofibrosis differs from the other myeloproliferative neoplasms because there isn't one myeloid cell line that proliferates more than the others.

In the other MPNs, the bone marrow produces large numbers of neoplastic, mature myeloid cells, with one cell line that predominates. That cell line spills out into the blood and can be seen there in large numbers. For example, in chronic myeloid leukemia, the neutrophil series is most prominent, whereas in polycythemia vera, it's the red blood cells that steal the show.

In contrast, in PMF, all the myeloid precursors (erythroid, neutrophilic, and megakaryocytic) proliferate wildly at first. But, as the disease progresses, something else happens: as the name suggests, the marrow becomes **fibrotic** (filled with collagen fibers).

Epidemiology

Primary myelofibrosis typically occurs in patients who are in their 60s or older, with a median age at presentation of 67 years. The annual incidence is approximately 1 to 2 cases per 100,000 people, similar to the other myeloproliferative neoplasms.

In primary myelofibrosis, what happens to the normal hematopoietic cells?

How Does a Patient With Primary Myelofibrosis Present (LO #1)?

Primary myelofibrosis, like the other myeloproliferative neoplasms, may initially be asymptomatic.

Symptoms

When there are symptoms of PMF, they may be vague, such as fatigue.

There may also be symptoms resulting from **splenomegaly**, such as a left upper quadrant fullness or discomfort, or early satiety (from the spleen impinging on the stomach). Splenomegaly is a feature of all the myeloproliferative neoplasms, but in PMF it's not just a slightly enlarged spleen—it's a spleen so large that its tip can be way down in the pelvis! This happens as the spleen tries to compensate for the lack of blood cell production in the bone marrow by generating these cells (in particular, RBCs) in the spleen (termed **extramedullary hematopoiesis**). This is rather like a factory worker trying to make up for absent coworkers. Extramedullary hematopoiesis also occurs in the liver, sk [illegible] where it is most clinically obvious.

Disease Progression

As bone marrow fibrosis worsens, bone marrow failure with resulting low blood cell production may result. The patient may then develop anemia-related fatigue and breathlessness, infections (because of low WBC count), and bleeding episodes (because of low platelet count).

Why does splenomegaly appear in PMF?

What Is the Pathophysiology of Primary Myelofibrosis (LO #2)?

The only known risk factor for PMF is ionizing radiation, and there is no recognized familial in [illegible] spontaneous gene mutations.

Genetics

Approximately 50-60% of PMF cases are associated with activating mutations in the ***JAK2* (Janus kinase 2) gene**. In *JAK2*-negative cases, there can be mutations in *CALR* (20-30% of cases) and *MPL* (5-10% of cases).

What gene is commonly mutated in primary myelofibrosis?

Pathology

Initially, PMF is characterized by bone marrow **proliferation of all myeloid cell lines**. So, the bone marrow is hypercellular and filled with excess neoplastic neutrophil precursors, red blood cell precursors, and megakaryocytes. Similar to ET, megakaryocytes in PMF are markedly increased and clustered with at [illegible]

Unlike the other myeloproliferative neoplasms, the bone marrow in PMF also has **extensive fibrosis**. The fibrosis is thought to be caused by fibroblast-stimulating cytokines released by megakaryocytes. Other myeloproliferative neoplasms can show a little fibrosis in the bone marrow, but it's nothing like the incredible amount of fibrosis seen in primary myelofibrosis, where the bone marrow eventually becomes nearly completely replaced by fibrosis, shown in the reticulin-stained fibers filling the bone marrow in Figure 1.

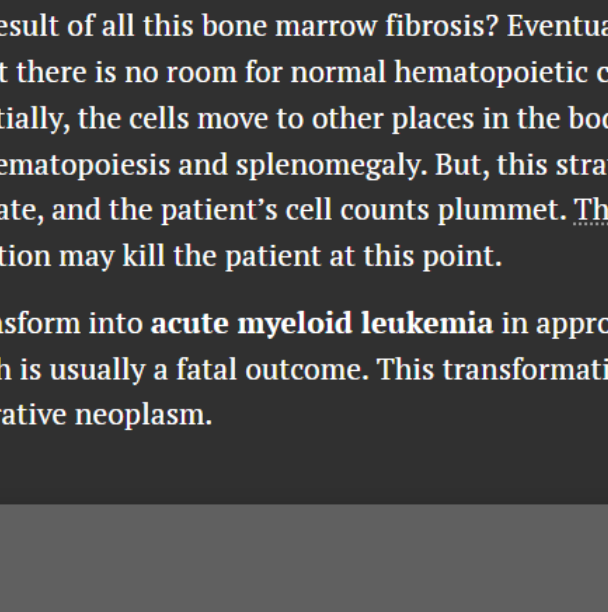

Figure 1

CLINICAL REASONING

Please note, the end [illegible] ntial thrombocythemia (E [illegible] ppearance, so it may be hard to distinguish these three neoplasms at this stage. Usually, having a prior history of one of these neoplasms is how we differentiate them.

Complications

What is the end result of all this bone marrow fibrosis? Eventually, there is so much fibrosis that there is no room for normal hematopoietic cells to proliferate, so initially, the cells move to other places in the body, causing extramedullary hematopoiesis and splenomegaly. But, this strategy eventually becomes inadequate, and the patient's cell counts plummet. Thrombosis, bleeding, or infection may kill the patient at this point.

PMF can also transform into **acute myeloid leukemia** in approximately 3% to 4% of cases, which is usually a fatal outcome. This transformation can occur in any myeloproliferative neoplasm.

What does the bone marrow look like in early primary myelofibrosis, and how does that change over time?

How Do We Diagnose Primary Myelofibrosis (LO #3)?

A diagnosis of PMF may be considered for an older patient with nonspecific symptoms of fatigue and fever, particularly if the spleen is massively enlarged. Diagnosis is therefore based on clinical findings and supported by diagnostic studies. We diagnose PMF using a complete blood count (CBC), peripheral blood smear and bone marrow exams, and molecular testing for genetic mutations.

Complete Blood Count

There typically is an elevated white blood cell count with a left shift and anemia, as is seen in the other myeloproliferative neoplasms, particularly chronic myeloid leukemia.

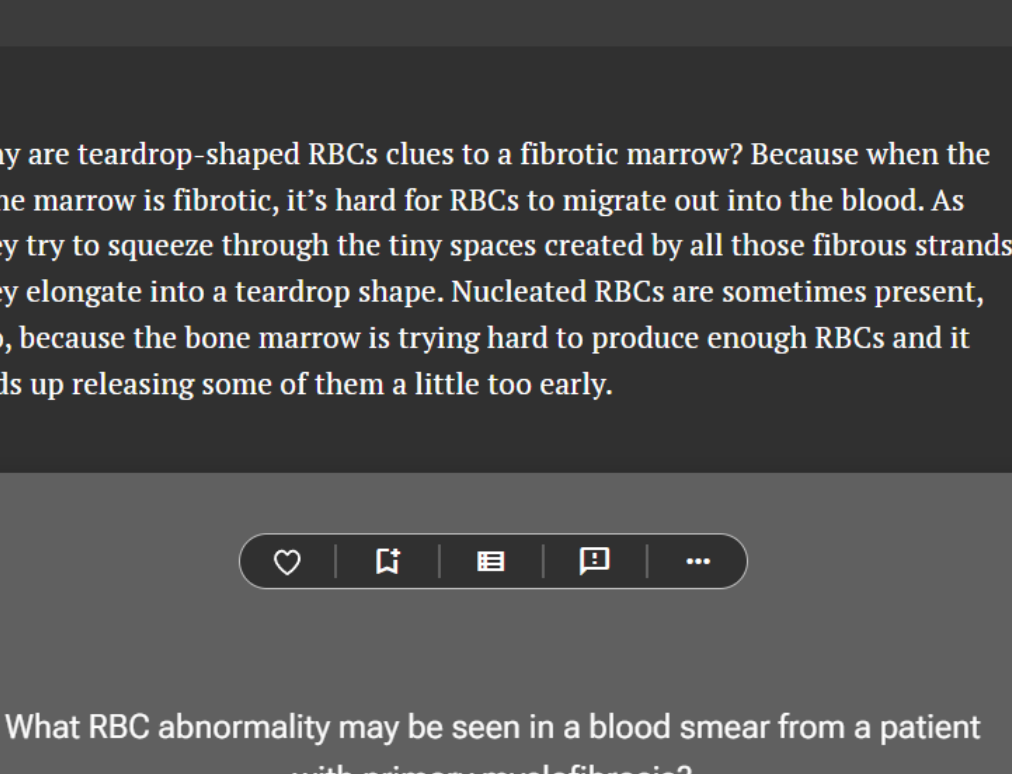

Figure 2. Teardrop red blood cells (black arrows) and a nucleated RBC (red arrow) are visualized in this peripheral smear.

CLINICAL CORRELATION

The typical appearance on the peripheral smear is referred to as a **leukoerythroblastic reaction**. This is where you see left-shifted granulocytic precursors (including rare blasts) and nucleated RBCs.

Why are teardrop-shaped RBCs clues to a fibrotic marrow? Because when the bone marrow is fibrotic, it's hard for RBCs to migrate out into the blood. As they try to squeeze through the tiny spaces created by all those fibrous strands, they elongate into a teardrop shape. Nucleated RBCs are sometimes present, too, because the bone marrow is trying hard to produce enough RBCs and it ends up releasing some of them a little too early.

What RBC abnormality may be seen in a blood smear from a patient with primary myelofibrosis?

Bone Marrow Biopsy

Because the CBC may suggest PMF but is not diagnostic of it, a bone marrow biopsy is needed to confirm the diagnosis. With all that fibrosis present, the bone marrow is usually difficult to aspirate. This is called getting a "dry tap." So, a core biopsy is usually more helpful. This will show either hypercellularity with fibrosis early in the disease or massive fibrosis with few cells in later stages.

Genetic Testing

Isolation of cells from the bone marrow enables genetic testing of these cells. Testing cells for specific aberrations such as the *JAK2* mutation may be useful, because it is present in approximately 50% of patients and is an important mutation in several myeloproliferative neoplasms.

How Do We Tre [illegible] LO #4)?

The treatment of primary myelofibrosis depends on disease severity. Patients without symptoms (i.e., low-risk asymptomatic) can be observed. Others require drug therapy or bone marrow transplantation.

Bone Marrow Transplantation

In more severe disease with poor prognosis, the nonfunctional bone marrow can be replaced. The procedure is called **allogeneic hematopoietic stem cell transplantation**, and it currently is the only therapy with curative potential for primary myelofibrosis. *In high- or intermediate-risk patients who are healthy enough to undergo it, a bone marrow transplant is recommended soon after diagnosis*.

An allogeneic transplant uses stem cells collected from the blood or bone marrow of a donor whose human leukocyte antigens (HLAs) are acceptable matches to the patient's.

Why not use the patient's own cells (autologous transplantation)?

Allogenic stem cell transplantation has dangerous potential side effects, such as **immunosuppression** with increased infections, as well as early **graft-vs-host disease**, in which transplanted cells attack the recipient. Acute graft-vs-host disease can cause skin rashes, cramping, diarrhea, nausea, vomiting, and elevation in liver function tests. The immunosuppressive drugs used after transplantation also have many potential adverse effects.

Low-Risk Treatment

Low-risk patients are either observed because they are asymptomatic, or treatment may be guided by their symptoms. This often includes monitoring and replacement, if needed, of platelet and RBC counts.

High-Risk Treatment

For patients who are high-risk, and not eligible for transplantation, we initiate drug therapy. There are two medications primarily used for PMF treatment, **ruxolitinib** and **hydroxyurea**.

As mentioned above, about 50% of PMF cases carry a *JAK2* mutation. This means that we can use the *JAK2* inhibitor ruxolitinib. As was discussed in the "Targeted Inhibitors Used in Cancer Therapy" brick, the JAK-STAT pathway is an important signaling pathway that regulates cell growth and proliferation. Ruxolitinib selectively inhibits *JAK1* and *JAK2* to mediate the dysregulated activity of the mutated *JAK2*. This can help with the proliferation and symptom (i.e., anemia and sple [illegible] F.

As with other targeted therapies, immunosuppression and graft-versus-host disease are concerns. Other concerns include pancytopenia, bruising, and hypercholesterolemia/ -triglyceridemia.

Hydroxyurea is an antimetabolite chemotherapy drug (Antimetabolite Brick) that can help reduce WBC counts in PMF. It does this by *interfering with DNA repair by inhibiting ribonucleotide reductase, halting the cell cycle at the G1/S phase*. Some of the adverse effects include cytopenia, mucocutaneous ulcers, diarrhea, and peripheral neuropathy. Additionally, it can cause teratogenicity in the fetus, so it should be used cautiously in women of child bearing age. Live vaccine administration is a contraindication while a patient is on hydroxyurea. In studies, hydroxyurea has shown a more favorable adverse effect profile, however, it may not be as effective as ruxolitinib.

CLINICAL CORRELATION

Momelotinib is a new JAK inhibitor that was recently approved by the FDA for PMF. Initial studies show that it is an effective treatment, and it may provide better anemia-related benefits over ruxolitinib.

Prognosis

The overall life expectancy for patients with this disorder is 6 years but varies depending on the severity of disease. Asymptomatic patients may live 15 years or more, whereas patients with massive bone marrow fibrosis, cytopenias, and symptoms may only li [illegible]

Which is currently the only therapy with curative potential for primary myelofibrosis?

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to VB, what is the next step in confirming this diagnosis? What do you tell VB about the condition?

Based on VB's presentation, physical examination, CBC, and peripheral blood smear, you strongly suspect primary myelofibrosis. You arrange for a bone marrow biopsy, and the results confirm your initial suspicion. You sit with VB and tell him that he has a blood disorder in which scar tissue replaces bone marrow, preventing normal blood cell formation. "But what's this pain on my left side?" VB asks. You explain that because the scar tissue makes the bone marrow unable to make normal blood cells, the spleen takes over and works hard and gets bigger to finish the job. You refer VB to a hematologist-oncologist who may recommend bone marrow transplantation.

Summary

- Primary myelofibrosis (PMF) is a myeloproliferative neoplasm characterized by a proliferation of all myeloid cells and, over time, increasing amounts of fibrotic tissue in the bone marrow.
- Signs and symptoms of PMF may include fatigue, anemia, and massive splenomegaly.
- Low platelet, WBCs, and RBC counts can cause symptoms of bleeding, infection, and fatigue in later stages.
- Most cases of PMF arise from spontaneous gene mutations.
- Approximately 50% of cases of primary myelofibrosis are associated with activating mutations in the *JAK2* gene, which are important for the Janus kinase/signal transducer and activator of transcription (JAK/STAT) signaling pathway to stimulate growth-promoting genes. *CALR* and *MPL* mutations comprise the majority of *JAK2*-negative cases.
- The bone marrow of patients with primary myelofibrosis has extensive fibrosis, caused by fibroblast-stimulating cytokines released by megakaryocytes.
- Splenomegaly is caused by the spleen trying to make up for the lack of bone marrow by undergoing extramedullary hematopoiesis.
- CBC and peripheral blood smear will reveal anemia, a left shift, and the presence of dacryocytes.
- Bone marrow biopsy will show varying degrees of fibrosis, as well as hypercellularity in early stages of the disease.
- Treatment includes medications (eg, JAK2 inhibitors, hydroxyurea) as well as allogenic bone [illegible] patients.
- Median survival is variable, averaging 6 years.

Review Questions

1. A 67-year-old male is admitted to the hospital for treatment of a lower-leg deep vein thrombosis using intravenous heparin. He has had 1 year of fatigue, bone pain, early satiety, and shortness of breath with exertion. A physical exam shows marked splenomegaly. His hemoglobin level on admission is 10.1 g/dL. A peripheral blood smear shows dacryocytes and nucleated RBCs. Which of the following is the best next step in management?

A. ○ Abdomen and pelvis CT scan
B. ● Bone marrow biopsy
C. ○ Discontinuation of the patient's heparin
D. ○ Starting the patient on myelosuppressive therapy
E. ○ Transfusion of packed red blood cells

Hide Explanation

The correct answer is bone marrow biopsy (B). The splenomegaly, anemia, and dacryocytes on blood smear suggest possible primary myelofibrosis. The next step is a bone marrow bio [illegible] oplasia. There is no reason for an abdomen [illegible] of the patient's heparin (C) is dangerous in the setting of deep vein thrombosis. Ruxolitinib or other myelosuppressive therapy (D) should not be administered before a diagnosis is made. The patient's current hemoglobin value does not warrant a transfusion of packed red blood cells (E).

2. Which of the following findings would most likely be seen on bone marrow biopsy in a patient with early primary myelofibrosis?

A. ○ Absence of reticulin and collagen within the bone marrow
B. ○ Clusters of high-grade dysplastic cells, surrounded by bone marrow fibrosis
C. ● Hypercellular bone marrow with an increased number of megakaryocytes
D. ○ Regression of bone marrow sinusoid
E. ○ Thinning of bony trabeculae with normocellular bone marrow

Hide Explanation

The correct answer is hypercellular bone marrow with an increased number of megakaryocytes (C). There will be an abundance, not absence of, reticulin and collagen deposition within the bone marrow (A) late in the disease. Although primary myelofibrosis will eventually cause fibrosis in the bone marrow, at no point in the disease process does it cause dysplasia of bone marrow cells (B). There would be no regression [illegible] of the disease (D), and the bone cortex [illegible] ary myelofibrosis (E).

3. Which of the following symptoms is most suggestive of extramedullary hematopoiesis in a patient with primary myelofibrosis?

A. ○ Bone aches
B. ○ Fatigue
C. ○ Fever
D. ● Splenomegaly
E. ○ Weight loss

Hide Explanation

The correct answer is marked splenomegaly (D). Patients with primary myelofibrosis typically have severe splenomegaly caused by RBC production there because RBCs cannot form in the fibrotic bone marrow. Extramedullary hematopoiesis can also cause abnormal liver function test results, but it would not cause bone aches (A), fatigue (B), fever (C), or weight loss (E). Fatigue is seen in myelofibrosis but is mainly caused by anemia, not the extramedullary hematopoiesis.

4. In what percentage of patients with primary myelofibrosis is a *JAK2* mutation implicate [illegible]

A. ○ 100%
B. ● 50% to 60%
C. ○ 10% to 20%
D. ○ 20% to 30%
E. ○ 0%

Hide Explanation

The correct answer is 50% to 60% (B). When learning about the myeloproliferative neoplasms, it is important to know what percentage of cases show the *JAK2* gene mutation because it is an important molecular correlate in clinical practice and can have implications for prognosis and treatment options. In essential thrombocythemia, 50% of cases exhibit a *JAK2* mutation. Almost 100% of polycythemia vera cases exhibit the mutation (A), and in chronic myeloid leukemia, none of the cases have the mutation (E). *JAK2* mutations in 10% to 20% (C) and 20% to 30% (D) are values unrelated to the myeloproliferative neoplasms covered here.

5. What is the primary mechanism of action of ruxolitinib in treating primary myelofibrosis (PMF)?

A. ○ Inhibits DNA polymerase to prevent cell replication
B. ○ Inhibits tyrosine kinase activity of BCR-ABL
C. ● Selectively inhibits JAK1 and JAK2 to regulate the JAK-STAT pathway
D. ○ Stimulates erythropoietin production to increase RBC counts

Hide Explanation

The correct answer is (C) Selectively inhibits JAK1 and JAK2 to regulate the JAK-STAT pathway. Ruxolitinib is a JAK1 and JAK2 inhibitor. It helps treat PMF by targeting the overactive JAK-STAT signaling pathway, which is commonly dysregulated due to a JAK2 mutation in ~50% of cases. This helps reduce splenomegaly and manage symptoms. Inhibiting DNA polymerase (A) is not the mechanism of ruxolitinib; this mechanism is more aligned with certain antiviral or chemotherapeutic agents. BCR-ABL tyrosine kinase inhibition (B) is the mechanism of drugs like imatinib, used in CML, not PMF. Ruxolitinib does not stimulate erythropoietin production (D); in fact, it can cause anemia and pancytopenia as adverse effects.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of primary myelofibrosis.~~

02 ☑ Explain [illegible] sis.

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of primary myelofibrosis.~~

02 ☑ ~~Explain the pathophysiology of primary myelofibrosis.~~

03 ☑ ~~Interpret clinical findings to properly diagnose primary myelofibrosis.~~

04 ☑ ~~Outline the management plan of primary myelofibrosis, including the rationale for bone marrow transplantation.~~

Objective Confidence: 4/4